

Breakout Strategy in MQL5

Overview

This code is an **Expert Advisor (EA)** written in MQL5 for the MetaTrader 5 trading platform. It implements a breakout strategy based on price range analysis and uses pending orders.

Objective

The EA implements a **breakout strategy** that:

- Calculates the price range based on historical bars.
- Places pending orders:
 - **Buy Stop** above the maximum price level + gap.
 - **Sell Stop** below the minimum price level - gap.
- Calculates Stop Loss (SL) and Take Profit (TP) levels using the ATR (Average True Range).
- Manages risk using the **Fixed Fractional** method, calculating trade volume based on a specified risk percentage.

Key Features

1. Breakout Levels Calculation:

- Analyzes the last N bars to determine the **maximum** (LivelloMAX) and **minimum** (LivelloMIN) price levels.

- Places:
 - **Buy Stop** above the maximum level + gap.
 - **Sell Stop** below the minimum level - gap.

2. Stop Loss and Take Profit Calculation:

- Based on ATR with configurable multipliers:
 - SL: Entry level $\pm$ (ATR $\times$ SL factor).
 - TP: Entry level $\pm$ (ATR $\times$ TP factor).

3. Risk Management:

- Uses the **Fixed Fractional** method to calculate trade volume based on the specified risk percentage and SL distance.

4. Conditions for Entry:

- Orders are placed only if:
 - There are no open positions.
 - There are no pending orders of the same type.

5. Cancellation of Pending Orders:

- Pending orders are canceled if:
 - The opposite order is executed.
 - A specified number of bars (**BarsValid**) has elapsed.

6. Time Filters:

- Trading is limited to the days of the week specified via input parameters.
- The code evaluates conditions only at the opening of a new bar.

Code Components

1. Input Variables:

- **MagicNumber**: Unique identifier for distinguishing EA trades.

- **RiskPercent:** Percentage of risk used for calculating trade volume.
- **BarreInEsame:** Number of bars to analyze for breakout levels.
- **GapEntry:** Gap distance from breakout levels for pending orders.
- **BarsValid:** Number of bars during which pending orders remain valid.
- **ATR-related:** Parameters for calculating SL and TP based on ATR.

2. Main Functions:

- **OnInit():**
 - Initializes global variables.
 - Configures the ATR indicator.
- **OnTick():**
 - Updates ATR data.
 - Cancels pending orders when appropriate.
 - Calculates breakout levels and places pending orders based on conditions.

3. Long and Short Trade Rules:

- ****Long**:**
 - Places a Buy Stop above the maximum level + gap.
 - SL below the entry level - (ATR $\times$ SL factor).
 - TP above the entry level + (ATR $\times$ TP factor).
- ****Short**:**
 - Places a Sell Stop below the minimum level - gap.
 - SL above the entry level + (ATR $\times$ SL factor).
 - TP below the entry level - (ATR $\times$ TP factor).

Usage

- **Instruments:** Works on currency pairs (Forex) and can be adapted for other instruments.

- **Timeframe:** Preferably H1 or higher.
- **Configuration:**
 - External parameters allow customization of risk, gap size, number of bars analyzed, etc.

Limitations

1. **Dependence on ATR:**

- The strategy heavily relies on ATR for calculating SL and TP, which may not be optimal in all markets.

2. **Backtesting:**

- Thorough testing on historical data is recommended to verify performance.

3. **Market Events:**

- The code does not account for macroeconomic events or sudden volatility spikes.